Multimodal Forecasting of fMRI Images from Audio–Video Data: Linear Model Analysis

Anastasiia German*
MIPT, Moscow, Russia
german.aia@phystech.edu

Daniil Dorin*
MIPT, Moscow, Russia
dorin.dd@phystech.edu

Abstract

We address the multimodal extension of linear fMRI forecasting, in which a sequence of voxel snapshots is predicted from both the video and the audio stream of a naturalistic film stimulus. We develop the approach in three stages of increasing methodological complexity. The starting point is a unimodal baseline that regresses the BOLD signal on Mel-Frequency Cepstral Coefficients of the soundtrack. We then introduce a banded ridge model that concatenates the audio embedding with a PCA-compressed video embedding and applies a separate ridge penalty to each modality; this analysis exposes the spatial heterogeneity of the audio-driven and the video-driven contributions to the BOLD signal but is limited by the architectural asymmetry between a low-dimensional handcrafted audio representation and a high-dimensional pretrained video representation. To remove this asymmetry, we replace the MFCC encoder with a self-supervised speech transformer of the same dimensionality as the video encoder and propose a per-voxel gated linear model that fits an independent ridge regression on each modality and combines the two predictions through a per-voxel scalar gate. The optimal gate is derived in closed form as a one-dimensional ordinary least-squares slope clipped to the unit interval. We further prove that under Gaussian regression noise the empirical distribution of the gate within the brain mask follows a censored Gaussian mixture, and validate the resulting model on the publicly available iEEG–fMRI dataset, where the gated model consistently improves over both single-modality baselines on every subject (Wilcoxon signed-rank with Holm correction).

Keywords: fMRI, BOLD signal, multimodal regression, ridge regression, censored Gaussian mixture.

1 Introduction

Functional magnetic resonance imaging (fMRI) measures changes in the blood-oxygen-level-dependent (BOLD) signal that follow the underlying neural activity with a delay of 4–8 seconds caused by the time required for the vascular system to respond to a local change in glucose demand [1]. The technique is the most widely used non-invasive tool for spatially-resolved measurement of stimulus-evoked brain activity and has been applied to a broad range of cognitive and clinical questions, including the study of autism and Alzheimer's disease [2].

In a naturalistic fMRI experiment the subject is exposed to a continuous external stimulus, in our case a short audio-visual film, and the BOLD time series is shaped by the visual and the auditory streams of the stimulus simultaneously. A previous study of one of the authors [3] considered the video stream as the only input and modelled each voxel by an independent linear regression. We extend that formulation to the multimodal case in which each voxel is

*Equal contribution.

associated with two simultaneously available stimulus streams that may carry partially redundant information about its dynamics; the central methodological problem is to allocate the per-voxel linear predictive capacity between the two streams in a principled way.

The auditory pathway is itself non-trivial. When sound enters the ear, hair cells in the inner ear convert mechanical movement into electrical signals that travel along the auditory nerve through the brainstem and the thalamus to the auditory cortex in Heschl's gyrus, with subsequent processing in adjacent regions that analyse pitch, speech patterns, musical tones, and rhythm [4–6]. To represent the audio stream in a fixed-dimensional form, two encoders have been used in successive stages of this work: a classical 15-dimensional Mel-Frequency Cepstral Coefficient encoder [7, 8], and a 768-dimensional self-supervised speech transformer (HuBERT [9]); the video stream is encoded by a Vision Transformer pretrained on ImageNet [10]. We use the publicly available iEEG–fMRI naturalistic stimulation dataset of [11], in which 30 subjects watched a short audio-visual film while the BOLD signal was recorded together with the time-aligned soundtrack.

Contributions.

- We document a sequence of three models of increasing methodological complexity on the dataset of [11]: a unimodal MFCC baseline, a banded ridge model concatenating audio with PCA-compressed video, and a per-voxel gated model with symmetric self-supervised audio and video encoders.
- For the banded ridge model we identify the regime (30 principal video components, separately tuned per-modality penalties) in which the multimodal forecast improves over the audio baseline by approximately +0.12% averaged over the brain mask and by +0.24% in the lateral-temporal region.
- We propose a closed-form gated linear method that fits an independent ridge regression on each modality and combines the two predictions through a per-voxel scalar gate $\alpha_{ijk} \in [0, 1]$. We derive the optimal gate analytically as a one-dimensional ordinary least-squares slope clipped to the unit interval and prove that, under Gaussian regression noise, the empirical distribution of the gate within the brain mask is a censored Gaussian mixture.
- We evaluate the gated model on nine subjects of [11] and report statistically significant improvements over both single-modality baselines on every subject (Friedman, Wilcoxon signed-rank with Holm correction, $\tilde{p} = 0.0039$; mean relative reduction of the brain-masked mean squared error +8.99% vs audio-only and +5.10% vs video-only).

Outline. Section 2 reviews prior work on linear encoding models for naturalistic stimuli. Section 3 fixes notation and the BOLD-prediction problem. Section 4 reports the MFCC unimodal baseline. Section 5 reports the banded ridge model with PCA-compressed video and its ROI analysis. Section 6 states the gated linear model with symmetric self-supervised encoders, derives the optimal gate, and proves the censored Gaussian-mixture result. Section 7 reports the numerical experiment on [11], and Section 9 concludes.

2 Related Work

Noise and signal in naturalistic fMRI. The BOLD signal is shaped not only by stimulus-driven activity but also by the brain's intrinsic "dark energy" — continuous default-mode-network activity that consumes the majority of neural energy and overlaps with stimulus-evoked signals [12]. Standard preprocessing techniques such as motion regression and physiological denoising can remove genuine neural variance together with nuisance noise, which has motivated careful and well-balanced denoising procedures [13]. Spontaneous BOLD fluctuations vary across regions and are constrained by structural connectivity [14], so general noise models may not capture local temporal differences. More recent topological methods, such as cubical persistence, identify stable features even during complex naturalistic tasks [15].

Linear encoding of naturalistic stimuli. Voxel-wise linear regression of the BOLD signal onto stimulus-derived features dates back to early task-based fMRI analyses [1] and was extended to long naturalistic recordings in subsequent work, in which short film clips serve as continuous probes of cortical organisation. Cortical maps obtained this way are typically interpreted in terms of modality, semantic category, or temporal selectivity [16]. The present work follows the linear-encoding tradition and adds an explicit per-voxel mixture between two pre-fit modality predictors.

Multimodal ridge regression. When two stimulus blocks of differing dimensionality are combined, naive concatenation favours the higher-dimensional block. Two remedies appear in the literature: block normalisation, dividing each block by $\sqrt{d_{\text{block}}}$ to equalise total variance, and *banded ridge regression* [17], in which a separate ridge penalty is applied to each block. The gated method introduced in this work differs from banded ridge: we fit each modality *independently* and combine the two predictions through a single per-voxel scalar, which avoids joint-fit hyperparameter coupling and admits a closed-form analytic optimum for the gate.

Audio and visual feature extraction. Deep neural networks optimised for auditory tasks reveal a hierarchical processing structure in the auditory cortex, mirroring the brain's organisation for processing speech and music [18–20], and Bayesian general linear models that allow per-voxel variability in noise level and temporal dependencies further improve the detection of genuine brain activity signals [21]. In this work, for the final model we use HuBERT [9], a transformer-based self-supervised speech encoder trained on 60K hours of unlabelled audio, complemented by an MFCC baseline [7] as a low-dimensional alternative. For the video stream we use per-frame embeddings produced by a Vision Transformer pretrained on ImageNet [10].

3 Preliminaries

3.1 Notation

Let $\nu_v \in \mathbb{R}_+$ denote the video frame rate, $\nu_a \in \mathbb{R}_+$ the audio sampling rate, and $\mu \in \mathbb{R}_+$ the fMRI sampling rate. The total duration of the stimulus is $t \in \mathbb{R}_+$. The video sequence is denoted by

$$\mathbf{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_{\nu_v t}], \quad \mathbf{p}_\ell \in \mathbb{R}^{W \times H \times C},$$

with width, height, and number of channels W, H, C. The audio waveform is denoted by

$$\mathbf{Q} = [q_1, \ldots, q_{\nu_a t}], \quad q_\tau \in \mathbb{R}.$$

The sequence of three-dimensional voxel snapshots is

$$\mathbf{S} = [\mathbf{s}_1, \ldots, \mathbf{s}_{\mu t}], \quad \mathbf{s}_\ell \in \mathbb{R}^{X \times Y \times Z},$$

where X, Y, Z are the dimensions of the voxel image. We write $\mathbf{s}_\ell = [v^\ell_{ijk}]$ with $v^\ell_{ijk} \in \mathbb{R}_+$. To reduce the running time, the snapshot may be precompressed by an averaging map $\boldsymbol{\chi} : \mathbb{R}^{X \times Y \times Z} \to \mathbb{R}^{X/c \times Y/c \times Z/c}$ with factor $c \in \{1, 2, 4, 8\}$; the result is kept under the same notation $X \times Y \times Z$.

3.2 Problem statement

Let $\Delta t \in \mathbb{R}_+$ be the hemodynamic delay. The forecasting problem is to construct a mapping $\mathbf{g}$ that predicts each fMRI snapshot from the preceding snapshot and from both stimulus streams sampled at the appropriately delayed time:

$$\mathbf{g}\big(\mathbf{p}_1, \ldots, \mathbf{p}_{k^v_\ell - \nu_v \Delta t};\, q_1, \ldots, q_{k^a_\ell - \nu_a \Delta t};\, \mathbf{s}_1, \ldots, \mathbf{s}_{\ell-1}\big) = \mathbf{s}_\ell, \quad \ell = 1, \ldots, \mu t, \tag{1}$$

where $k^v_\ell = \ell\nu_v/\mu$ and $k^a_\ell = \ell\nu_a/\mu$ are the video and audio time indices that correspond to the ℓ-th fMRI snapshot.

We assume the Markov property on the sequence of snapshots,

$$\mathbf{g}(\mathbf{p}_{k^v_\ell - \nu_v \Delta t}, q_{k^a_\ell - \nu_a \Delta t}) = \boldsymbol{\delta}_\ell = \mathbf{s}_\ell - \mathbf{s}_{\ell-1} = [\delta^\ell_{ijk}], \quad \ell = 2, \ldots, \mu t,$$

which reduces (1) to predicting the per-voxel difference $\boldsymbol{\delta}_\ell$ of two consecutive snapshots from the appropriately delayed stimulus.

3.3 Brain mask for evaluation

Background voxels with negligible BOLD signal would dominate the average mean squared error if included; we therefore restrict every reported error to a per-subject brain mask $\mathcal{M}$. The mask is constructed in two steps. The temporal mean of the four-dimensional BOLD volume is thresholded by Otsu's method [22] to drop air and out-of-brain background voxels. Within the resulting mask, voxels in the lowest decile by temporal variance are removed to drop static-noise voxels at the skull rim. The same mask is used for all three estimators introduced in Sections 4–6 so that the comparison is performed on a common set of voxels.

4 MFCC unimodal baseline

The first stage of the analysis fits an audio-only linear model with MFCC features and characterises its hemodynamic delay and weight distribution; the model is later used as a baseline against which the multimodal extensions are compared.

4.1 Feature extraction

The audio stream is mapped to a per-frame Mel-Frequency Cepstral Coefficient (MFCC) vector $\mathbf{x}^a_\ell \in \mathbb{R}^{d_a}$ with $d_a = 15$. The continuous audio signal is divided into short overlapping frames, each frame is transformed by the short-time Fourier transform, the resulting frequency spectrum is passed through a bank of triangular filters spaced along the mel scale, the band energies are compressed by a logarithm, and the discrete cosine transform is applied to the log-energy values to reduce redundancy [7, 8].

4.2 Voxel-wise linear regression

For each voxel (i, j, k), the regression model is linear,

$$f_{ijk}(\mathbf{x}^a_\ell, \mathbf{w}_{ijk}) = \langle \mathbf{x}^a_\ell, \mathbf{w}_{ijk} \rangle, \qquad \mathbf{w}_{ijk} \in \mathbb{R}^{d_a},$$

fitted by minimisation of the regularised squared loss

$$\mathcal{L}_{ijk}(\mathbf{w}_{ijk}) = \sum_{\ell=2}^{N} \big(f_{ijk}(\mathbf{x}^a_\ell, \mathbf{w}_{ijk}) - \delta^\ell_{ijk}\big)^2 + \lambda \,\|\mathbf{w}_{ijk}\|_2^2.$$

The minimiser is the closed-form ridge solution $\widehat{\mathbf{w}}_{ijk} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \boldsymbol{\Delta}_{ijk}$, where $\mathbf{X} \in \mathbb{R}^{(N-1)\times d_a}$ is the audio design matrix and $\boldsymbol{\Delta}_{ijk}$ is the per-voxel target vector of delta-snapshots.

4.3 Results

The MFCC baseline was evaluated on three representative subjects (7th, 22nd, 31st) with voxel values normalised to $[0, 1]$. The brain-masked mean squared error of the recovered snapshot is of the order of 10^{-5} in all three subjects, indicating that the audio stimulus indeed carries linearly extractable information about the BOLD signal. The distribution of the fitted weight vector $\widehat{\mathbf{w}}_{ijk}$ is non-degenerate and approximately symmetric around zero, with no concentration around a single value: the model picks up a broad range of spectral features rather than being driven by one or two coefficients. A sweep over the hemodynamic delay $\Delta t \in \{0, \dots, 20\}$ seconds on the auditory cortex region only reveals a clear global minimum at approximately $\Delta t \approx 5$–8 seconds, consistent with the established range of the hemodynamic response function [1].

5 Banded ridge with PCA-compressed video

The MFCC baseline of Section 4 uses the audio stream only. The natural next step is to add the video stream and ask whether the multimodal forecast improves over the audio baseline. A naive concatenation $[\mathbf{X}^v \,|\, \mathbf{X}^a]$ of the 768-dimensional ViT video embedding with the 15-dimensional MFCC audio embedding overweights the video block; the standard remedy is *banded ridge regression*, in which a separate ridge penalty is applied to each block.

5.1 Model

Let $\mathbf{X}^v \in \mathbb{R}^{(N-1)\times d_v}$ and $\mathbf{X}^a \in \mathbb{R}^{(N-1)\times d_a}$ be the design matrices of the two modalities. The banded ridge problem for voxel (i, j, k) is

$$\min_{(\boldsymbol{\theta}_v, \boldsymbol{\theta}_a)} \left\| \boldsymbol{\Delta}_{ijk} - [\mathbf{X}^v \,|\, \mathbf{X}^a] \begin{bmatrix} \boldsymbol{\theta}_v \\ \boldsymbol{\theta}_a \end{bmatrix} \right\|_2^2 + \lambda_v \|\boldsymbol{\theta}_v\|_2^2 + \lambda_a \|\boldsymbol{\theta}_a\|_2^2, \tag{2}$$

where the per-modality penalties $\lambda_v, \lambda_a \in \mathbb{R}_+$ are tuned independently by grid search on the training fold. The solution is closed-form.

5.2 Compressing the video stream

The video embedding from a Vision Transformer pretrained on ImageNet [10] has dimension $d_v = 768$, much larger than $N - 1$ test residuals available per voxel. Without compression, the video block dominates the joint fit even with banded penalties. We therefore apply principal component analysis to the video block before concatenation, keeping the first d_v^{PCA} principal components on the training fold and projecting the test fold onto the same basis. PCA is applied to the *video* block only; the audio block remains at $d_a = 15$.

5.3 Sweep over the number of video principal components

A sweep over $d_v^{\mathrm{PCA}} \in \{5, 10, 15, 20, 30, 50, 100\}$ on a representative subject identifies $d_v^{\mathrm{PCA}} = 30$ as the unique value at which the multimodal model improves over the audio baseline. Values $d_v^{\mathrm{PCA}} \geq 50$ overfit to noisy components, and $d_v^{\mathrm{PCA}} < 20$ retains insufficient information. With $d_v^{\mathrm{PCA}} = 30$, $\lambda_v = 10^3$, $\lambda_a = 10^4$ (selected by grid search), and the hemodynamic delay $\Delta t = 8$ seconds, the multimodal model reaches MSE $= 2.805 \cdot 10^{-5}$ against the audio baseline MSE $= 2.808 \cdot 10^{-5}$, a relative reduction of approximately $+0.12\%$ averaged over the brain mask. The multimodal curve is below the audio curve for every $\Delta t \geq 4$ seconds, and both curves share the same global minimum at $\Delta t = 8$ seconds.

5.4 ROI analysis

The whole-brain effect of adding the video stream is small but not uniformly distributed. We partition the brain mask into approximate geometric regions of interest (ROI) and compute the relative reduction of the brain-masked mean squared error in each ROI separately:

Table 1: ROI-level relative reduction of the brain-masked mean squared error of the banded ridge multimodal model with $d_v^{\mathrm{PCA}} = 30$ over the MFCC audio baseline, on a representative subject.

ROI	# voxels	ΔMSE	Improved voxels
Lateral-temporal	880	+0.244%	57.7%
Posterior	1920	−0.028%	51.4%
Anterior	1920	−0.221%	50.4%
Whole brain	17,790	+0.124%	51.7%

The largest relative reduction is observed in the lateral-temporal region, consistent with the intuition that the audio-visual stimulus drives multimodal integration in the auditory cortex and adjacent regions. The anterior region exhibits a slight negative effect: adding the video block degrades the audio-only baseline there, indicating that the joint fit allocates predictive capacity to a stream that does not carry relevant information in that ROI.

5.5 Additional video feature variants

Two additional video feature variants were tested with the banded ridge model and the same evaluation protocol:

- **Temporal derivatives of ViT features.** The frame-to-frame difference of the 768-dimensional ViT embedding may better capture stimulus transitions than the raw embedding, since the BOLD signal reflects changes in neural activity rather than absolute levels. Concatenating the embedding with its temporal derivative did not improve over the embedding alone.

- **Low-level video features.** A 6-dimensional hand-crafted vector of frame brightness, contrast, and motion energy was used in place of the ViT embedding. These features may better describe what early visual cortex processes, but the resulting multimodal model did not outperform the MFCC baseline on the whole brain.

Both variants confirm that the modest whole-brain improvement of the banded ridge model is not an artefact of the specific video encoder choice but a structural property of the model: the per-modality penalties shape the joint fit globally and cannot, by construction, allocate the predictive capacity differently to different voxels. This observation motivates the per-voxel gated model of Section 6.

6 Gated multimodal model

The banded ridge analysis of Section 5 establishes two facts. The audio-driven and the video-driven contributions to the BOLD signal are spatially heterogeneous (Section 5.4). A single global pair (λ_v, λ_a) cannot allocate the per-voxel predictive capacity differently to auditory-cortex voxels and to visual-cortex voxels and consequently leaves most of the heterogeneity unmodelled. In this section we propose a model that addresses the heterogeneity directly with a per-voxel scalar gate.

6.1 Symmetric self-supervised encoders

The 15-dimensional MFCC representation of Section 4 and the 768-dimensional ViT video embedding of Section 5 differ in dimensionality by a factor of 50, which biases the joint fit even when the dimensionality mismatch is partly absorbed by PCA and banded penalties. To remove the asymmetry, we replace MFCC by HuBERT [9], a self-supervised speech transformer with $d_a = 768$, so that both modalities are represented by 768-dimensional embeddings produced by a transformer backbone. Both feature streams are resampled onto a common time grid that matches the video frame rate ν_v and standardised per dimension on the training fold.

For every fMRI snapshot index ℓ, two paired samples are obtained per voxel,

$$\mathfrak{D}^v_{ijk} = \{(\mathbf{x}^v_{k^v_\ell},\, \delta^\ell_{ijk}) \mid \ell = 2, \dots, N\}, \qquad \mathfrak{D}^a_{ijk} = \{(\mathbf{x}^a_{k^a_\ell},\, \delta^\ell_{ijk}) \mid \ell = 2, \dots, N\},$$

where $N = \mu(t - \Delta t)$ is the total number of fMRI snapshots used for training and testing.

6.2 Voxel-wise single-modality ridge models

For each voxel (i, j, k) and each modality $m \in \{v, a\}$, the regression model is linear, fitted by minimisation of the regularised squared loss with a modality-specific regularization parameter $\lambda_m \in \mathbb{R}_+$. The minimiser is the closed-form ridge solution

$$\widehat{\mathbf{w}}^m_{ijk} = \left(\mathbf{X}^{m\top}\mathbf{X}^m + \lambda_m \mathbf{I}\right)^{-1} \mathbf{X}^{m\top}\, \boldsymbol{\Delta}_{ijk}, \tag{3}$$

where $\mathbf{X}^m \in \mathbb{R}^{(N-1)\times d_m}$ is the design matrix of modality m. The ridge penalty is denoted by λ_m so that α remains reserved for the per-voxel modality gate introduced below.

6.3 Per-voxel modality gate

Let $\widehat{\boldsymbol{\delta}}^v_{ijk}, \widehat{\boldsymbol{\delta}}^a_{ijk} \in \mathbb{R}^{N-1}$ be the time series of the per-voxel video-only and audio-only delta predictions obtained from (3). The proposed multimodal predictor at voxel (i, j, k) is the convex combination

$$\widehat{\boldsymbol{\delta}}_{ijk}(\alpha_{ijk}) = \alpha_{ijk}\, \widehat{\boldsymbol{\delta}}^v_{ijk} + (1 - \alpha_{ijk})\, \widehat{\boldsymbol{\delta}}^a_{ijk}, \qquad \alpha_{ijk} \in [0, 1], \tag{4}$$

where α_{ijk} is the per-voxel *modality gate*: $\alpha_{ijk} \to 1$ selects the video-only prediction and $\alpha_{ijk} \to 0$ selects the audio-only prediction.

Proposition 1. *Define the audio-only residual and the modality-prediction difference at voxel (i, j, k) as*

$$\mathbf{r}_{ijk} := \boldsymbol{\Delta}_{ijk} - \widehat{\boldsymbol{\delta}}^a_{ijk}, \qquad \mathbf{d}_{ijk} := \widehat{\boldsymbol{\delta}}^v_{ijk} - \widehat{\boldsymbol{\delta}}^a_{ijk}.$$

The value of α_{ijk} that minimises the squared training error of (4) *on the constraint set* $[0, 1]$ *is*

$$\widehat{\alpha}_{ijk} = \operatorname{clip}\left(\frac{\langle \mathbf{d}_{ijk},\, \mathbf{r}_{ijk}\rangle}{\|\mathbf{d}_{ijk}\|_2^2 + \varepsilon},\ 0,\ 1\right), \tag{5}$$

where $\varepsilon > 0$ is a small numerical floor preventing division by zero when $\widehat{\boldsymbol{\delta}}^v_{ijk} \equiv \widehat{\boldsymbol{\delta}}^a_{ijk}$.

Proof. Substitution of (4) into the squared training error gives

$$\mathcal{L}_{ijk}(\alpha) = \sum_{\ell=2}^{N} \left(\delta^\ell_{ijk} - \alpha\, \widehat{\delta}^{v,\ell}_{ijk} - (1 - \alpha)\, \widehat{\delta}^{a,\ell}_{ijk}\right)^2 = \|\mathbf{r}_{ijk} - \alpha\, \mathbf{d}_{ijk}\|_2^2.$$

The objective is a strictly convex one-dimensional quadratic in α with leading coefficient $\|\mathbf{d}_{ijk}\|_2^2 \geq 0$. Setting $\mathrm{d}\mathcal{L}_{ijk}/\mathrm{d}\alpha = 0$ yields $\widehat{\alpha}^*_{ijk} = \langle \mathbf{d}_{ijk}, \mathbf{r}_{ijk}\rangle / \|\mathbf{d}_{ijk}\|_2^2$. Since $[0, 1]$ is convex, the constrained minimiser is the projection of $\widehat{\alpha}^*_{ijk}$ onto $[0, 1]$, which coincides with the clip operator. The numerical floor ε is required only when $\|\mathbf{d}_{ijk}\|_2 = 0$, in which case any value of α achieves the minimum and the regularised solution returns 0. □

The estimator (5) is the slope of a one-dimensional ordinary least-squares regression of $\mathbf{r}_{ijk}$ on $\mathbf{d}_{ijk}$ truncated to $[0, 1]$.

6.4 Reconstruction formula

Let $\widehat{\mathbf{W}}^v, \widehat{\mathbf{W}}^a$ be the matrices obtained by stacking the per-voxel weight vectors $\widehat{\mathbf{w}}^v_{ijk}, \widehat{\mathbf{w}}^a_{ijk}$ along the rows, and let $\widehat{\boldsymbol{\alpha}} \in \mathbb{R}^{XYZ}$ be the vector of per-voxel gates from (5). The vectorised forecasted snapshot is

$$\widehat{\mathbf{s}}^R_\ell = \mathbf{s}^R_{\ell-1} + \widehat{\boldsymbol{\alpha}} \odot \widehat{\mathbf{W}}^v \mathbf{x}^v_\ell + (\mathbf{1} - \widehat{\boldsymbol{\alpha}}) \odot \widehat{\mathbf{W}}^a \mathbf{x}^a_\ell, \tag{6}$$

where $\odot$ denotes the elementwise product. The full estimator consists of three closed-form solves – one ridge solve per modality through (3) and one scalar per voxel through (5) – and adds exactly one parameter per voxel beyond the two single-modality models.

7 Numerical experiment

7.1 Dataset

The dataset [11] contains the recordings of 63 subjects of which 30 underwent fMRI scanning. Among the latter, 16 are male and 14 are female, with ages ranging from 7 to 47 years and a mean age of 22 years. Each subject watched a 390-second audio-visual film while the BOLD signal was recorded by a 3 T scanner. The dataset additionally provides the soundtrack of the film at 44.1 kHz mono, time-aligned with the fMRI recording. Table 2 summarises the recording characteristics.

Table 2: Dataset description.

Quantity	Notation	Value
Duration of examination	t	390 s
Video frame rate	ν_v	25 Hz
Audio sampling rate	ν_a	44.1 kHz
fMRI frame rate	μ	1.64 Hz
Video frame dimensions	W, H, C	$640, 480, 3$
fMRI volume dimensions	X, Y, Z	$40, 64, 64$

7.2 Experimental setup

The recordings are split along the time axis into a training fold and a test fold in proportion 70/30 without shuffling. The video frames and the audio waveform are resampled onto a common time grid at the video frame rate ν_v and standardised per dimension on the training fold. The voxel values of the BOLD volume are normalised to $[0, 1]$ by the MinMax procedure fitted on the training fold. To reduce the running time, the BOLD volume is precompressed by an AvgPool3D operator with stride and kernel of size $c = 2$. The video features are produced by the 768-dimensional pre-classification embedding of a pretrained image transformer [10]; the audio features by HuBERT [9] (final-layer 768-dimensional embedding). The hemodynamic delay is fixed at $\Delta t = 4$ seconds for the gated model and $\Delta t = 8$ seconds for the banded ridge model, in each case at the minimum of the corresponding Δt sweep.

7.3 Compared estimators

Three estimators are compared on the same train/test split and the same hemodynamic delay:

- the *audio-only* model, the voxel-wise ridge (3) on $\mathbf{X}^a$ alone with penalty λ_a;
- the *video-only* model, the voxel-wise ridge (3) on $\mathbf{X}^v$ alone with penalty λ_v;

- the *gated multimodal* model, the per-voxel-gated combination (4) of the two single-modality predictors with $\widehat{\alpha}_{ijk}$ given by Proposition 1.

For all three estimators the test mean squared error is averaged only over the brain mask of Section 3.3.

7.4 Subject-level results

The three estimators were fitted on nine subjects (IDs 04, 07, 09, 11, 14, 15, 16, 22, 31) with $\lambda_v = \lambda_a = 200$ and $\Delta t = 4$ seconds. On every subject, the gated model has the lowest brain-masked test mean squared error, with the audio-only baseline as the worst of the three on every subject and the video-only baseline as the middle estimator on every subject. The mean relative reduction of the brain-masked test mean squared error of the gated model over the two single-modality baselines is +8.99% over audio-only and +5.10% over video-only.

7.5 Statistical significance

We follow the testing scheme of Demšar [23] for the comparison of multiple models over multiple related samples: a group test on the three estimators, pairwise tests for the two comparisons of interest, and a family-wise correction.

Friedman test The Friedman test on the three estimators rejects the global null hypothesis of equality at the $\alpha = 0.05$ level: $\chi^2(2) = 16.22$, $p = 0.0003$. Pairwise tests are therefore justified.

Wilcoxon signed-rank test for related samples For each baseline, the one-sided Wilcoxon signed-rank test for related samples with the alternative $\mathrm{MSE}_{\mathrm{gated}} < \mathrm{MSE}_{\mathrm{baseline}}$ assigns the minimum achievable rank statistic $W = 0$, $p_{\mathrm{raw}} = 0.0020$ (the lower bound of the one-sided Wilcoxon distribution at $n = 9$, achieved when all nine signed differences agree in sign).

Holm correction The two raw p-values are corrected for family-wise error rate by the Holm step-down method on a family of two tests: $\tilde{p} = 0.0039$ for both comparisons. The corrected p-values are below the standard threshold $\alpha = 0.05$, and the null hypothesis of no difference is rejected for both comparisons. Table 3 summarises the result.

Table 3: Pairwise statistical comparison of the gated multimodal model against the two single-modality baselines on nine subjects, with Holm correction for the family-wise error rate. $\tilde{p}$ denotes the Holm-corrected p-value. ΔMSE is the mean relative reduction of the brain-masked test mean squared error.

Comparison	$\tilde{p}$ (Holm)	ΔMSE, %
Gated vs Audio-only	0.0039	+8.99
Gated vs Video-only	0.0039	+5.10

The dependence of the brain-masked test mean squared error on Δt on the same grid $\{0, 1, \ldots, 20\}$ seconds confirms that the ordering of the three estimators is preserved across all delays at which the BOLD signal is predictable from the stimulus.

7.6 Censored Gaussian-mixture fit of the per-voxel gate

The empirical distribution of $\widehat{\alpha}_{ijk}$ on the brain-mask voxels is fitted to the censored Gaussian mixture (**??**) of Proposition **??** with $K = 2$. The boundary masses $\widehat{p}_0$ and $\widehat{p}_1$ are identified with the empirical fractions of voxels at $\widehat{\alpha}_{ijk} < 10^{-3}$ and $\widehat{\alpha}_{ijk} > 1 - 10^{-3}$ respectively. The mixture weights $(\widehat{\pi}_1, \widehat{\pi}_2)$ and the per-component truncated-Gaussian parameters $(\widehat{\mu}_k, \widehat{s}_k)$ are estimated by expectation-maximisation on the continuous component. The fit is tight: $D \approx 0.007$ and $W_1 \approx 0.001$ on a representative subject, both well below the conventional thresholds $D < 0.05$ and $W_1 < 0.01$.

7.7 Spatial layout of the per-voxel gate

The per-voxel gate $\widehat{\alpha}_{ijk}$ is reshaped into a three-dimensional volume on the downsampled BOLD grid and overlaid in colour onto the temporal mean of the BOLD volume rendered in grayscale. Voxels outside the brain mask of Section 3.3 are rendered transparent. Red corresponds to $\widehat{\alpha}_{ijk} \to 1$ (video-driven voxel) and blue to $\widehat{\alpha}_{ijk} \to 0$ (audio-driven voxel). Per-voxel improvement maps of the gated model over the audio-only baseline show concentrated positive improvement in the visual cortex; improvement maps over the video-only baseline show concentrated positive improvement in the auditory cortex.

8 Discussion

The progression of the three models reported in this work — the MFCC unimodal baseline of Section 4, the banded ridge model with PCA-compressed video of Section 5, and the gated multimodal model with symmetric self-supervised encoders of Section 6 — illustrates the value of two methodological choices for naturalistic-stimulus fMRI forecasting. Symmetric high-dimensional encoders for the two modalities remove the architectural asymmetry that biases the joint fit of a banded ridge model. Per-voxel mixture between two independently fitted single-modality predictors models the spatial heterogeneity of the audio-driven and the video-driven contributions to the BOLD signal directly, instead of relying on a single global pair of penalties.

Two limitations of the present work deserve to be highlighted. The sample size of nine subjects places the minimum achievable one-sided p-value of the Wilcoxon signed-rank test at $2^{-9} \approx 0.002$, which is the smallest possible signal of statistical evidence at this sample size and not a property of the data. Replication on the full sample of thirty subjects, or on an independent dataset, would tighten the inference. The linear model class limits the prediction quality on voxels driven by non-linear interactions between modalities; a neural-network-based extension of the gated mixture, in which the per-voxel gate is replaced by a learned function of the local feature vector, is a natural next step.

9 Conclusion

A linear method for the multimodal forecasting of fMRI snapshots from audio-video stimuli has been developed in three stages: a unimodal MFCC baseline, a banded ridge model concatenating audio with PCA-compressed video, and a per-voxel gated model with symmetric HuBERT and ViT encoders. The gated model fits two voxel-wise ridge regressions on the two modality streams independently and combines the two modality-specific predictions through a per-voxel scalar gate $\alpha_{ijk} \in [0, 1]$. The gate is the closed-form minimiser of the per-voxel mean squared error and admits the analytic form of a one-dimensional ordinary least-squares slope truncated to the unit interval (Proposition 1). Under Gaussian regression noise, the empirical distribution of the gate within the brain mask follows a censored Gaussian mixture (Proposition **??**), validated empirically with Kolmogorov–Smirnov effect size $D \approx 0.007$ and Wasserstein-1 distance $W_1 \approx 0.001$. The gated model significantly improves over both single-modality baselines on every subject in our nine-subject sample (Friedman, Wilcoxon signed-rank with Holm correction, $\tilde{p} = 0.0039$; mean relative reduction of the brain-masked mean squared error $+8.99\%$ versus audio-only and $+5.10\%$ versus video-only).

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